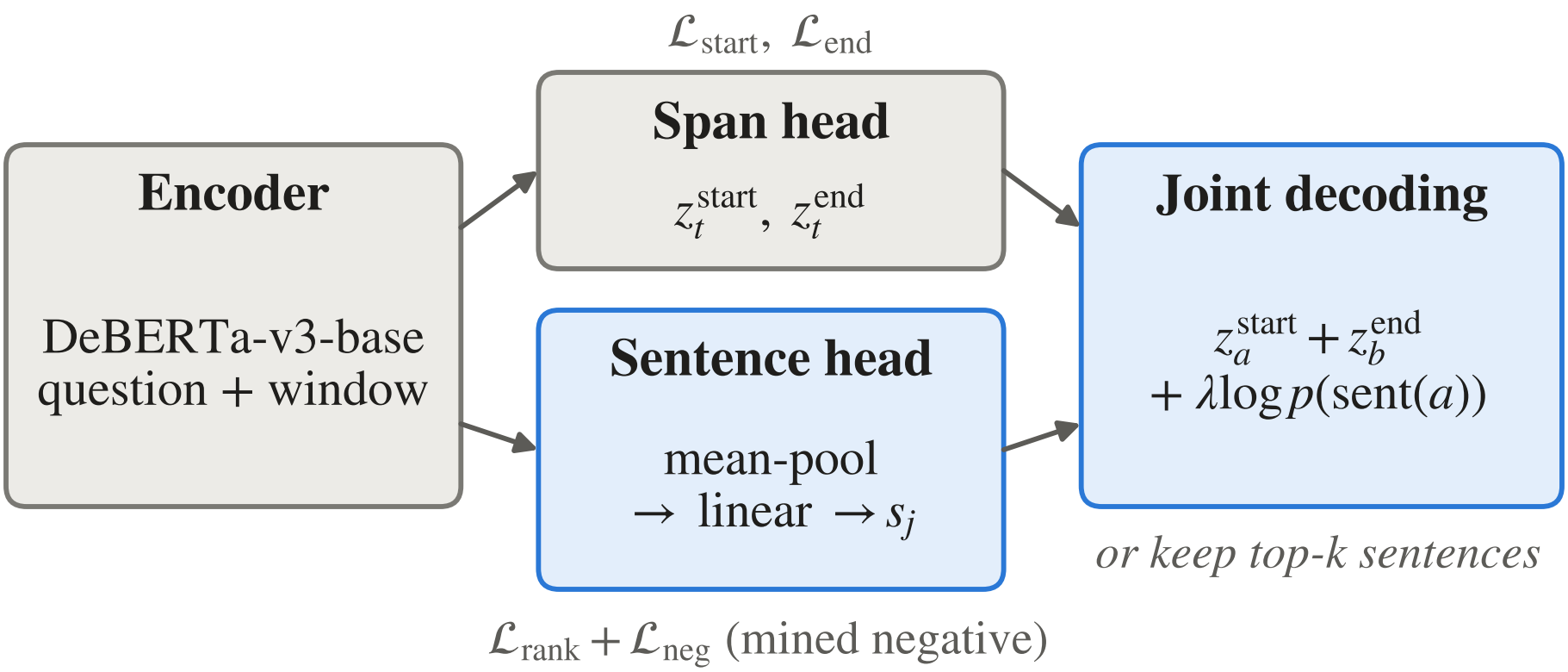
$\mathcal{L}_{\text{start}}$, $\mathcal{L}_{\text{end}}$
Span head
z_t^{start}, z_t^{end}
Encoder
DeBERTa-v3-base
question + window
Sentence head
mean-pool
→ linear → s_j
Joint decoding
$z_a^{\text{start}} + z_b^{\text{end}}$
$+ \lambda \log p(\text{sent}(a))$
or keep top-k sentences
$\mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{neg}}$ (mined negative)